Week 2 – Literature Survey, Referencing Tools and Experimental Skills (Lean Notes)

Lectures covered: L6 (overview of literature survey), L7 (Web of Science), L8 (Scopus), L9 (wrap-up), L10 (BibTeX with LaTeX), L11 (Word with bibliographic sources), L12 (Word endnotes), L13 (experimental skills). ⭐ = tested in the Week 2 assignment.

1. Concept cards (in the order taught)

1. Literature survey is not Googling

A literature survey = survey of **authoritative technical literature** (universities, research labs), not private websites. Closest thing in Google is **Google Scholar**.

2. Why do a literature survey ⭐

- No one can repeat the whole history of a topic: learn **what others did, which approaches, where the gaps are, and what you add**.
- **Novelty in familiarity** (new technique, familiar area) and **familiarity in novelty** (familiar technique, new area) both need a survey to know what is familiar.
- **Avoid duplication** (a research degree needs original work; unintentional duplication is also not acceptable), **clarify controversial results**, learn the **state of the art**, set the context, spot anomalies.
- Do it at the start **and throughout**.

3. Where to find literature

- **Open access** ⭐: freely available (readers download free). Limited but growing. Peer review, author fees and reputation vary, so only "free to readers" is always true.
- **INDEST** (MHRD programme, coordinated by IIT Delhi; institutes in tiers) for government-funded institutes; **your library** (ask the librarian); **peer sharing**: ResearchGate, academia.edu.

4. Types of sources ⭐

Level	Examples
Primary (original reports)	periodicals/journals, research reports, conference proceedings, official publications, standards, theses and dissertations, forthcoming papers
Secondary	compilations, reviews, reference books, handbooks, textbooks, monographs, abstracting services
Tertiary	dictionaries, year books, bibliographies, lists of secondary sources

5. Identifiers ⭐

- **ISBN** = book. **ISSN** = journal/newspaper/proceedings series (identifies the *journal*, not one article).
- **DOI** (Digital Object Identifier) = one online document; the DOI agency maps it to the current URL, so it keeps working when the file moves. **A DOI alone locates one article** (volume, page or ISSN alone cannot).
- **UDC** = catalogue number to find a book physically. Bibliographic file formats: **RIS** and **BIB (BibTeX)**.
- Archival (permanent, traceable) references: full journal citation, DOI, book with ISBN. Not: an unread reference, a website, an article missing year/ISSN.

6. Measures of impact ⭐

- **Journal impact factor:** how often a journal's articles are cited. Compare journals only **within the same area** (active areas have higher values); relative, controversial.
- **Citation count:** times one article is cited; very high = **citation classic**.
- **h-index** (author): an author has h-index **n** if **n papers are each cited at least n times**. Combines output and impact. Example: 5 papers cited at least 5 times each, no sixth such paper, means h = 5.

7. Search strategies and databases ⭐

- **Keyword search:** topic words, author name, or the paper's own keywords; use variants.
- **Backward chronological search:** read a paper's **reference list**, then theirs, and so on, back in time → reaches the **earliest contributor**.
- **Forward chronological search:** pick a key paper, then find papers that **cite it** (needs a citation database); can only go up to the present.
- **Abstracting databases** (subscription): INSPEC (physics, electrical, computing), COMPENDEX (engineering), PubMed (medicine), MathSciNet, Chemical Abstracts. **Citation databases:** Web of Science (Thomson ISI) and Scopus (Elsevier). Wikipedia is not one.

8. Publishers and alerts

- About **80%** of journals come from a few publishers: **Elsevier and Springer** (majority), Taylor & Francis, Nature Publishing, Maney; plus societies (APS, AIP). Portals: **ScienceDirect** (Elsevier), **SpringerLink**.
- **RSS/content alerts** push new articles to your inbox; special-interest sites (e.g. imechanica.org).

9. Reference styles and managers

- Order can be numeric (order of citation), chronological or alphabetical; the style depends on university/publisher. A **reference manager** switches style. **JabRef** is free and open source (others are paid; check your library).

10. Before using Web of Science or Scopus

- Ask the librarian: do we subscribe? access by **IP address** or **roaming login**? **proxy** username/password?
- **Register with your official email** so searches and history are saved. For Web of Science also register on **EndNote Web** (myendnoteweb.com) and keep both in two browser tabs. Keep EndNote's **"Unfiled" list empty** before collecting, so two surveys don't mix.

11. Running a keyword search (demo: "melt spinning" OR "rapid solidification")

- Find **alternative keywords** and join them with **OR** (union of results).
- Choose the **right field** (Topic, Title, Author, DOI, Year…); a wrong field gives too few hits. Scopus fields: article title, abstract, keywords, author, affiliation.
- **Time span:** all years the first time; later only the new year. Scopus: keep **all subject areas** open to catch interdisciplinary work.
- Hits: about 18,000 (Web of Science) and 22,000 (Scopus) = **mature area**, so sort and pick a few.

12. Sorting the results (try every type, pick the top few of each)

Relevance · **date, newest first** (latest work) · **date, oldest first** (how the area began) · **times cited, high to low** (citation classics) · times cited, low to high (ignored but maybe useful) · **usage count** (Web of Science only: how many users picked it, last 6 months or all time) · first author · source/conference title.

13. Collecting and exporting references

- **Web of Science:** tick items → **Add to Marked List** (like a shopping cart) → 3 steps: select records, select content (include **abstract**; skip times cited/usage count), destination **EndNote Online**. In EndNote: Unfiled → select all → **Add to Group** (new name) → Format → Export References → style **BibTeX** → Save (rename the file to **.bib**). Web of Science cannot give BibTeX directly.

- **Scopus:** tick → **Save to list** → open list → select all → **Export** → BibTeX with citations and abstract → file (rename meaningfully). Scopus exports directly (RIS, CSV for Excel, BibTeX, text).
- Why BibTeX: native format of many managers, convertible to others.

14. The .bib file and JabRef

- A **.bib** file is plain text: each entry starts with **@article** (type) and fields in braces; you can cut and paste in Notepad to merge or split. TeXmaker/TeXworks (MiKTeX) show it neatly.
- **JabRef** shows it like a **spreadsheet** (rows = papers, columns = fields): search, sort, add notes, read abstracts, split/merge files, export formats, work **offline**.

15. Wrap-up: why manage references this way

- Collect **full texts** (ScienceDirect, SpringerLink, ResearchGate, library) **and** the reference list, and link them with the manager.
- An article has about **10–30** references, a thesis **100 or more**: typing by hand causes typos; every cited paper must be listed; style and sequence must follow the target journal. LaTeX and Word do this automatically. Same works for figures and tables.

16. BibTeX with LaTeX (L9–L10)

- Keep the **.bib** file in the same folder as the **.tex** file; each entry needs a **unique BibTeX key** (JabRef can generate keys).
- In the .tex file: `\bibliography{name}` (name without .bib), `\bibliographystyle{unsrt}` (numbered in order of citation; other styles: alpha, name-year), and `\cite{key}` anywhere.
- Compile order: **LaTeX → BibTeX → LaTeX → LaTeX** (then pdfLaTeX). Move a `\cite` and the numbers re-sequence automatically.

17. Word with bibliographic sources (L9, L11)

- JabRef: **File → Export → Microsoft Office 2007 XML**. In Word: **References → Manage Sources → Browse** (pick the XML) → copy all to **Current List** → **Insert Citation** → at the end **Bibliography → References**.
- Finally **delete unused sources** in Manage Sources and **Update Citations**.

18. Word endnote entries (L9, L12)

- **References → Insert Endnote**, then type or paste the reference (e.g. from JabRef). Numbers re-sequence when you move text.
- Fix looks: endnote numbers come in **Roman**, change to **Arabic** in the Footnote and Endnote dialog; use **Style Inspector** to remove the superscript; in **Draft view → Show Notes → Endnote Separator** delete the line. **Cross-reference** re-cites a note. Manual typing is more error-prone than XML sources.

19. Theory vs experiment (L13)

- Both ask: *if one quantity changes, what happens to another?* **Theory** (equations) **predicts**; **controlled experiments** (hold conditions constant, change one quantity) **test**. Experiments are "supreme": a theory whose predictions fail is **incomplete**, however elegant.
- But only if the experiment is **run correctly**. Be ready to defend how you ran it.
- **Experimental skill** = tear your experiment apart and ask at every step *"what can go wrong here?"*, then **calibrate, cross-check and compensate**.

20. Temperature and pressure

- A furnace display shows the temperature **at the thermocouple**, not at your sample (often **5–10 cm** away, with gas flow cooling it). Display 800 °C can mean sample 750–760 °C, and the error differs at each setpoint (you may be at 475, 750, 1000 °C when you think 500, 800, 1100): data are correct but not for the temperature you claim.

- Fix: add a **second thermocouple** (K-type) just **above the sample**, **calibrate** it with a standard, and **raise the setpoint** (e.g. 840 °C if the sample then reads 800 °C). For gas atmospheres use a controlled inlet, an outlet and a **bubbler** so there is flow but no pressure build-up.
- **Pressure:** same ideas: check where the sensor sits, that it is not clogged or dirty, and verify against a standard.

21. Humidity

- **Relative humidity (RH)** = moisture in the gas as a % of the maximum it can carry at that temperature (maximum rises with temperature; steam tables give only the *upper limit*).
- A humidifier at a set temperature does **not** automatically give 100% RH (design, flow rate, efficiency). Sensors can be fooled by condensation, so **check independently**: pass the gas through a **desiccant box**, **weigh it before and after**; weight gain = water carried by a known gas volume; compare with theory.
- Adjust: e.g. run the bottle at 77 °C so that, after cooling to 70 °C, the gas is 100% RH at 70 °C. **Heat the line** (heat tracing) so water stays **vapour**, not liquid drops. Handle gases like hydrogen safely.

22. Current and voltage

- **Cross-check** the display with another instrument; a **clamp-on meter** reads current without breaking the circuit (display 50 A, true 47 A is possible).
- **Voltage location matters:** a **two-probe** measurement includes contact resistances; a **four-probe** measurement (current through outer leads, voltmeter on inner leads) removes them.

$$R_{\text{two-probe}} = R_c + R_s + R_c \qquad R_{\text{four-probe}} \approx R_s \qquad R = \frac{V}{I}$$

Terms: Rc = contact resistance of one lead on the sample; Rs = resistance of the sample itself; V = voltage between the voltmeter leads; I = current through the sample.

What it is for: shows why two-probe readings are too high and why four-probe gives the sample's own resistance (and from it, resistivity, which accounts for sample size).

Tiny number: Rc = 1 Ω each, Rs = 10 Ω → two-probe reads 12 Ω (20% high); four-probe reads about 10 Ω, because the voltmeter lead carries almost no current (contact drop 3–4 orders of magnitude smaller).

2. Top traps (most likely first)

1. **DOI vs ISSN/volume/page:** only the DOI is unique to one article; ISSN names the journal.
2. **Backward search** (reference lists) finds the **earliest** work; **forward/citation search** finds later work. Sorting by year or citations does not.
3. **Open access** only guarantees free reading, not "no peer review", "no fees" or "poor quality".
4. **Primary vs secondary vs tertiary:** reviews and compilations are secondary; year books and directories are reference aids.
5. **Display temperature is not sample temperature**; the sample is usually colder, so raise the setpoint after measuring.
6. **Two-probe overestimates** sample resistance; four-probe corrects it.
7. **Humidifier setpoint ≠ delivered RH**; line must stay hot; verify with desiccant weighing.
8. Experiment vs theory: data **sometimes** confirm the model; experiments are supreme only if correctly run; a mismatch can be an error, a model limit or a new effect.
9. Impact factors are **not comparable across fields**; h-index = n papers with at least n citations each.
10. Web of Science exports BibTeX only **through EndNote Web** (Unfiled list must be empty first); Scopus exports directly. Use **OR** between keyword variants.

11. LaTeX compile order is LaTeX, BibTeX, LaTeX, LaTeX; Word needs the **XML** version of the .bib; delete unused sources.

3. Assignment questions with solutions

Q1. Experimental data generated from a setup must validate the theoretical predictions or models designed for that setup. *(Q1)* - A. True B. False C. Sometimes D. Never

Solution (card 19): Experiments **test** a theory; they are not forced to agree. They agree in some cases and not in others (setup error, model limits, or a new effect). **Answer: C.**

Q2. Literature survey is important because (select all that apply) *(Q2)* - A. It gives a summary of work done till date in an area - B. It ensures research work is not repeated - C. It helps understand approaches taken and the gaps - D. It helps clarify controversial results

Solution (card 2): All four are reasons given in the lecture. **Answer: A, B, C and D.**

Q3. Which method of searching is suitable to identify the earliest contributor of a scientific outcome? *(Q3)* - A. Keyword search, then sort by year B. Citation search, then sort by year - C. Keyword search, then sort by citations D. Back-reference search following a keyword search

Solution (card 7): Keyword + sort by year only gives the oldest *matching* paper. Citation search goes *forward* in time. Sorting by citations finds influential, not earliest. Back-referencing goes through reference lists back to the origin. **Answer: D.**

Q4. Which bibliographic detail alone is sufficient and unique to locate an article in an archival journal? *(Q4)* - A. ISSN B. DOI C. Page number D. Volume number

Solution (card 5): ISSN = journal; page and volume mean something only with journal and year. DOI = one article. **Answer: B.**

Q5. Which is always true for an open access journal? *(Q5)* - A. Not peer reviewed B. Articles are freely available for download C. They charge for publication D. They are of dubious reputation

Solution (card 3): Only free availability is part of the definition. **Answer: B.**

Q6. Which is NOT a recommended practice to choose publications for the references of an article? *(Q6)* - A. Relevant publications actually referred to B. Latest publications by the editor of the journal, because he/she is the editor - C. Original publications on the article's topic D. Own publications for further details

Solution: (outside the lectures) References should be works you used, the original sources and your own related work. Citing an editor only because of the editor's position is padding. The question asks for the bad practice. **Answer: B.**

Q7. Which index measures the impact of an author in his/her specialisation? *(Q7)* - A. Total number of articles B. Hirsch index C. Views of the latest article D. Downloads of all articles in a year

Solution (card 6): h-index combines number of papers and their citations. **Answer: B.**

Q8. Primary sources for a literature survey (select all that apply) *(Q8)* - A. Periodicals and research reports B. Conference proceedings and official publications - C. Compilations and reviews D. Directories and year books

Solution (card 4): A and B are original reports. C is secondary, D is reference/tertiary. **Answer: A and B.**

Q9. Abstracting databases for systematic manual searches include (select all that apply) *(Q9)* - A. INSPEC and COMPENDEX B. PubMed C. Wikipedia D. Chemical Abstracts service

Solution (card 7): A, B, D are abstracting databases; Wikipedia is an encyclopedia. **Answer: A, B and D.**

Q10. Scenario: six resources were collected: (1) a reference in a journal article that you have not read yet; (2) a link to a professional society's website; (3) a published article with journal, year, volume and page number; (4) an article from a new journal with volume and page number but year and ISSN still due; (5) a newly accepted journal article with a DOI link; (6) a book with an ISBN. **Which are archival references for a publication?** *(Q10)* - A. Only 1, 2, 3 B. Only 3, 5, 6 C. Only 3, 4, 6 D. Only 1, 2, 5

Solution (card 5): Archival = permanent and traceable. (3) full citation, (5) DOI, (6) ISBN book qualify. (1) unread and unverified, (2) websites change, (4) missing year and ISSN. **Answer: B (3, 5 and 6).**

Answer key: Q1 C · Q2 A,B,C,D · Q3 D · Q4 B · Q5 B · Q6 B · Q7 B · Q8 A,B · Q9 A,B,D · Q10 B

Outside the lectures: only Q6 (why citing an editor's papers is poor practice) and the reasons for rejecting items 1, 2 and 4 in Q10 are not directly taught.